

Question Number	Answer	Mark
5(a)(i)	The only correct answer is D xylem <i>A is not correct because magnesium ions are transported in the xylem</i> <i>B is not correct because plasmodesmata is not a tissue</i> <i>C is not correct because magnesium ions are transported in the xylem</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(a)(ii)	An explanation that makes reference to the following points: - magnesium (ions) are needed for the formation of {chlorophyll / chloroplasts} (1) (yellow leaves are due to) {fewer / no} {chlorophyll (molecules) / chloroplasts / green pigments} (1) (reduced growth is due to) {reduced / no} {production of glucose / carbohydrate / photosynthesis} (1) need clear statement	accept magnesium (ions) are needed for {activation of enzymes / protein synthesis} less Mg=less chlorophyll = mp1&2 less chlorophyll linked to magnesium = mp1&2 accept fewer activated enzymes accept {no/reduced} {metabolic reactions / respiration} if linked to the mp1AG only	(3)

Question Number	Answer
*5(b) 1&2 3 4 5 6 7 8	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • there is (a greater) increase in mean leaf area with eggshells / increase is exponential with eggshells • similar increase in mean leaf area from day 10-17 (for vine plants with and without eggshells) • there is (a greater) increase in {mean shoot length / growth} with eggshells • similar increase in mean shoot length from day 10-25 (for vine plants with and without eggshells) • there is a greater chlorophyll content with eggshells • no data from leaf area / shoot length beyond {52-55} days • relevant comment on {error/SD} bars e.g. overlap between xxx and xxx therefore no significant difference / no overlap between xxx and xxx therefore significant difference / correct comment about error bar size linked to {reliability / validity} • (because crushed eggshells contain) calcium (ions) needed to form calcium pectate • calcium pectate {joins adjacent cells together / involved in forming middle lamella} • greater leaf area results in more {chlorophyll / chloroplasts/ light absorption} • greater leaf area results in increased photosynthesis • more {chlorophyll / light absorption} results in increased photosynthesis • increased {shoot/growth} results in more {light absorption / photosynthesis} • more photosynthesis results in more growth • more {photosynthesis / chlorophyll} results in more {glucose production / chemical energy store} • more glucose results in more growth • {glucose / chemical energy store} can be used in {respiration to provide ATP} / ATP needed for mitosis • glucose can be converted into {amino acids / glycerol} • increased protein synthesis would result in increased {growth / fruit production/chlorophyll production} only need increased once • (increased) calcium pectate would result in (increased) {growth of {plant / leaves} / support / strength / stability} • cost of adding eggshells / comparison of cost with traditional fertiliser • eggshells {more sustainable (as using waste product) / renewable resource / biodegradable} / deter pests such as slugs

(6)			
			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	description of data 1 mark – description of at least one graph 2 marks – description of at least two graphs
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Level 1 plus linkages of chlorophyll content/leaf area/shoot length to light absorption or photosynthesis plus linkage of calcium ions to calcium pectate
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Level 2 plus consideration of the benefits of increased {glucose/protein/calcium pectate} to the plant and the cost involved / other benefits of eggshells